

Lecture 2

From now on unless otherwise noted,

$$G = \mathrm{SL}_2(\mathbb{F}_q)$$

$\mathbb{F}_q$ = fin field with q elements

$\uparrow$
pf

$$\text{Let } B := \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} : a \in \mathbb{F}_q^\times, b \in \mathbb{F}_q \right\}$$

$$U := \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} : b \in \mathbb{F}_q \right\}$$

$$T = \left\{ \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} : a \in \mathbb{F}_q^\times \right\}$$

$$\text{So } B = T \ltimes U$$

$$d: \mathbb{F}_q^\times \xrightarrow{\sim} T$$

$$a \mapsto \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}$$

$$u: \mathbb{F}_q \rightarrow U$$

$$x \mapsto \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$$

$$\text{So } d(a)u(x)d(a)^{-1} = u(a^2x)$$

Set $s = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, it satisfies $s^2 = -I_2$ and

$$s d(a) s^{-1} = d(a^{-1})$$

s normalizes T , so setting $N = \langle T, s \rangle$, we have
 $T \triangleleft N$ and $N/T \cong \mathbb{Z}/2\mathbb{Z}$

Prop (Bruhat Decomposition) $G = B \sqcup B s B = B \sqcup U s B$

Proof Note $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} = \dots = \begin{pmatrix} x & x \\ a & x \end{pmatrix}$ so $B \cap U s B = \emptyset$

And if $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$ has $c \neq 0$, then

$$s^{-1} u(-\frac{a}{c}) g = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -\frac{a}{c} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \dots = \begin{pmatrix} x & x \\ 0 & x \end{pmatrix} \in B$$

$$\Rightarrow g \in U s B. \quad \square$$

We call T the (a) split torus. We now define the non-split torus. Let $\mathbb{F}_{q^2} = \text{deg } 2 \text{ ext of } \mathbb{F}_q$.

Fixing an $\mathbb{F}_q$ -basis for $\mathbb{F}_{q^2} \Rightarrow d: GL_{\mathbb{F}_q}(\mathbb{F}_{q^2}) \xrightarrow{\sim} GL_2(\mathbb{F}_q)$

Since $\mathbb{F}_{q^2}^\times \subseteq GL_{\mathbb{F}_q}(\mathbb{F}_{q^2})$

$$\Rightarrow d': \mathbb{F}_{q^2}^\times \hookrightarrow GL_2(\mathbb{F}_q)$$

For $\alpha \in \mathbb{F}_{q^2}^\times$,

$$\text{Tr}(\alpha) = \text{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(\alpha) = \text{Tr} \circ d'(\alpha) = \alpha + \alpha^q$$

$$\text{Nm}(\alpha) = \text{Nm}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(\alpha) = \det \circ d'(\alpha) = \alpha^{q+1}$$

Let $\mu_n = n^{\text{th}}$ roots of unity in an algebraic closure of $\mathbb{F}_q$

Then $\mathbb{F}_{q^2}^\times = \mu_{q^2-1} \xrightarrow{d'} GL_2(\mathbb{F}_q)$ Set $T' = d'(\mu_{q+1})$,
 $\mu_{q+1} \xrightarrow{d'} G = SL_2(\mathbb{F}_q)$ the non-split torus

Since the Frob $F: \mathbb{F}_{q^2} \rightarrow \mathbb{F}_{q^2}$ is $\mathbb{F}_q$ -linear, $F \in GL_{\mathbb{F}_q}(\mathbb{F}_{q^2})$,
 $x \mapsto x^q$

$\tilde{s}' = d'(F) \in GL_2(\mathbb{F}_q)$ satisfies

(a) $\tilde{s}'^2 = I_2$ and

(b) $\tilde{s}' d'(\alpha) \tilde{s}'^{-1} = d'(\alpha^q) \quad \forall \alpha \in \mathbb{F}_{q^2}^\times$

(a)+(b) $\Rightarrow$ min poly of $\tilde{s}'$ is $X^2 - 1$, in part $\det(\tilde{s}') = -1$

Take $\alpha_0 \in \mathbb{F}_{q^2}^\times$ with $\text{Nm}(\alpha_0) = -1$. Set $s' = \tilde{s}' d'(\alpha_0)$

Then $s' \in G$ and

(a) $s'^2 = \tilde{s}' d'(\alpha_0) \tilde{s}' d'(\alpha_0) = \tilde{s}' d'(\alpha_0^{q+1}) \tilde{s}' = -I_2$

(b) $s' d'(\alpha) s'^{-1} = d'(\alpha^{q+1}) \quad \forall \alpha \in \mathbb{F}_{q^2}^\times$

Setting $N' = \langle T', s' \rangle$, $T' \triangleleft N'$ and $N'/T' \cong \mathbb{Z}/2\mathbb{Z}$.

$$\text{Set } Z = \{ \pm I_2 \} \cong \begin{cases} \mathbb{Z}/2\mathbb{Z} & \text{if } p > 2 \\ 1 & \text{if } p = 2 \end{cases}$$

Thm 1 If $q > 3$, then $D(G) := [G, G] = G$

and Z is the only nontrivial normal subgroup. In particular $G/Z = PSL_2(\mathbb{F}_q)$ is simple.

2. If $q = 2$, $G \cong S_3$, $D(G) = T' \cong A_3$ and $G = T' \rtimes U$.

3. If $q = 3$, $D(G) = N'$ and the nontrivial normal subs are

Z and N' . Moreover $G = N' \rtimes U$ and $G/Z \cong A_4$.

Lemma 1 $G = \langle U, s U s^{-1} \rangle = \langle U, s \rangle$

Proof $H := \langle U, s U s^{-1} \rangle \subseteq \langle U, s \rangle \subseteq G$. By the Bruhat Decomp, $G = B \sqcup B s B$ and $B = T \rtimes U$, it suffices to show $s \in H$ and $T \subseteq H$. A computation shows

$s = u(-1) [s u(-1) s^{-1}] u(-1) \in H$
 and for $a \in \mathbb{F}_q^\times$

$$d(a) = u(a) [s u(a^{-1}) s^{-1}] u(a) s \in H \quad \square$$

Lemma 2 $\bigcap_{g \in G} g B g^{-1} = Z$.

Proof Set $H = \bigcap_{g \in G} g B g^{-1}$. Clearly $Z \subseteq H$.

Now $B n s B s^{-1} = T$, and $u = u(1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, one checks

$$T n u T u^{-1} = Z. \text{ So}$$

$$Z \subseteq H \subseteq B n s B s^{-1} n u B u^{-1} n u s B s^{-1} u^{-1} = Z. \quad \square$$

Proof of Thm, part 1 We assume $q > 3$. Then $\exists \alpha \in \mathbb{F}_q^\times$ with $\alpha^2 \neq 1$. Then $d(a) u(x) d(a)^{-1} u(x)^{-1} = u(\alpha^2 - 1)x$

$$\Rightarrow U \subseteq D(G) \Rightarrow D(G) = G \text{ by Lemma 1.}$$

Now let H be a nontrivial normal subgroup.